

FORT PROVIDENCE, NT, Canada

WMO: 710870

Lat: 61.317N

Lon: 117.602W

Elev: 162

StdP: 99.40

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.4	(c) -36.5	(d) -43.7	(e) 0.1	(f) -39.2	(g) -40.8	(h) 0.1	(i) -36.3	(j) 5.5	(k) -15.3	(l) 4.7	(m) -18.3	(n) 0.6	(o) 80	(p) 0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	28.7	17.0	26.8	16.4	25.0	15.6	18.4	26.0	17.5	24.6	16.5	23.1	2.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.3	20.5	14.6	10.6	19.7	13.7	9.9	19.0	52.7	26.2	49.9	24.7	47.0	23.3	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.1	(b) 5.2	(c) 4.4	(e) -41.7	(f) 31.9	(g) 2.1	(h) 1.7	(i) -43.2	(j) 33.1	(k) -44.5	(l) 34.1	(m) -45.6	(n) 35.0	(o) -47.2	(p) 36.2		
			DB													
			WB													
				(e) -41.8	(f) 20.0	(g) 2.1	(h) 1.2	(i) -43.3	(j) 20.9	(k) -44.5	(l) 21.6	(m) -45.6	(n) 22.2	(o) -47.1	(p) 23.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-23.0	-19.2	-13.6	-1.9	7.1	14.1	17.1	14.7	8.8	0.0	-12.1	-20.5	(6)	
(7)		DBStd	15.42	7.74	6.91	8.31	6.77	4.98	3.99	3.27	3.69	4.38	4.63	6.42	7.73	(7)	
(8)		HDD10.0	5053	1022	818	732	356	114	10	1	6	74	310	664	946	(8)	
(9)		HDD18.3	7568	1281	1052	990	606	347	135	62	123	287	567	914	1204	(9)	
(10)		CDD10.0	569	0	0	0	1	26	133	221	150	38	1	0	0	(10)	
(11)		CDD18.3	43	0	0	0	0	1	8	24	9	1	0	0	0	(11)	
(12)		CDH23.3	772	0	0	0	0	32	189	368	169	14	0	0	0	(12)	
(13)		CDH26.7	178	0	0	0	0	5	43	91	37	2	0	0	0	(13)	
(14)	Wind	WSAvg	1.9	1.3	1.5	1.8	2.2	2.4	2.1	2.0	2.1	2.2	2.0	1.7	1.3	(14)	
(15)	Precipitation	PrecAvg	356	18	15	15	14	27	50	61	52	35	26	22	20	(15)	
(16)		PrecMax	453	34	26	30	23	47	97	107	92	73	44	44	34	(16)	
(17)		PrecMin	245	6	6	2	6	6	7	29	9	11	8	8	10	(17)	
(18)		PrecStd	48	7	6	7	5	11	19	20	18	14	9	8	6	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	3.5	10.4	17.6	27.0	30.5	31.8	30.3	25.3	16.7	6.6	0.5	(19)	
(20)			MCWB	-1.5	0.6	4.4	8.4	14.0	17.0	18.6	18.3	15.4	10.4	2.6	-1.4	(20)	
(21)		2%	DB	-5.4	-2.0	6.1	13.8	23.2	27.7	29.0	27.3	21.8	11.5	0.6	-4.0	(21)	
(22)			MCWB	-6.1	-3.4	1.9	6.7	12.3	15.9	17.6	17.0	13.5	6.8	-0.4	-4.5	(22)	
(23)		5%	DB	-9.6	-6.4	2.6	11.2	19.9	25.4	27.1	25.1	19.3	8.6	-1.6	-7.2	(23)	
(24)			MCWB	-10.0	-7.4	-0.3	5.2	10.9	15.1	16.9	16.0	12.2	5.4	-2.2	-7.6	(24)	
(25)		10%	DB	-12.6	-9.4	-0.8	8.6	16.9	23.3	25.3	22.9	16.7	6.2	-3.5	-9.5	(25)	
(26)			MCWB	-13.0	-10.1	-2.7	3.7	9.2	14.1	16.3	15.2	11.3	4.2	-4.1	-9.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	0.8	4.5	8.7	14.6	18.5	20.2	19.5	16.5	10.5	2.4	-1.3	(27)	
(28)			MCDWB	0.7	3.4	9.8	16.6	25.1	27.1	28.5	27.5	23.5	15.9	5.8	0.3	(28)	
(29)		2%	WB	-6.0	-3.6	2.0	6.9	12.8	17.1	18.7	18.0	14.4	7.2	-0.2	-4.6	(29)	
(30)			MCDWB	-5.2	-2.0	5.7	13.3	22.4	25.0	26.1	25.3	20.0	10.4	0.5	-3.9	(30)	
(31)		5%	WB	-10.0	-7.4	-0.2	5.5	11.2	16.1	17.9	16.8	13.0	5.7	-2.2	-7.6	(31)	
(32)			MCDWB	-9.7	-6.4	2.5	10.8	19.4	23.4	25.2	23.1	18.0	8.2	-1.5	-7.2	(32)	
(33)		10%	WB	-13.0	-10.1	-2.7	3.9	9.7	14.9	17.0	15.7	11.7	4.2	-4.1	-9.9	(33)	
(34)			MCDWB	-12.6	-9.4	-0.7	8.3	16.4	21.7	23.6	21.7	16.4	6.2	-3.6	-9.5	(34)	
(35)	Mean Daily Temperature Range			MDBR	8.2	10.9	13.9	13.0	13.1	13.9	12.9	12.3	11.1	7.2	7.2	7.5	(35)
(36)		5% DB	MCDBR	11.2	14.0	15.8	15.6	18.5	17.8	16.2	16.5	15.6	12.0	8.7	9.8	(36)	
(37)			MCWBR	10.6	12.4	11.9	9.6	9.6	8.2	7.0	7.9	8.3	7.8	7.6	9.4	(37)	
(38)		5% WB	MCDWB	11.2	13.8	15.3	14.8	17.3	15.2	14.5	14.5	14.2	11.2	8.5	9.8	(38)	
(39)			MCWBR	10.6	12.3	11.5	9.3	9.2	7.5	6.9	7.4	8.1	7.5	7.5	9.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.198	0.243	0.267	0.321	0.346	0.353	0.379	0.353	0.309	0.275	0.201	0.158	(40)	
(41)		taud		2.120	2.272	2.278	2.274	2.430	2.431	2.380	2.485	2.560	2.523	2.174	1.964	(41)	
(42)		Ebn at Noon		571	767	866	872	878	874	837	835	821	733	561	441	(42)	
(43)		Edh at Noon		35	61	86	107	100	102	105	88	68	49	32	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg		0.35	1.24	3.03	5.01	5.71	6.10	5.47	4.25	2.71	1.15	0.37	0.18	(44)	
(45)		RadStd		0.02	0.08	0.19	0.33	0.41	0.53	0.34	0.25	0.26	0.12	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.62	N/A	N/A	N/A	N/A	N/A	+18
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page